

Multiplying Two Polynomials with NTT - Full Worked Example

This document does one complete polynomial multiplication using the NTT method, done entirely by hand-style arithmetic, following the exact same structure as FIPS 203's real algorithms (Algorithm 9-12), just shrunk down to $n=8$ instead of $n=256$ so every number can be seen and checked by hand.

Sizes used for this example: $n=8$, $q=17$, $\text{zeta}=9$ (instead of the real $n=256$, $q=3329$, $\text{zeta}=17$). $\text{zeta}=9$ was chosen because it has the same special property real Kyber's zeta needs: $\text{zeta}^4 = -1 \pmod{17}$, same as $\text{zeta}^{128} = -1 \pmod{3329}$ in the real thing.

What we are multiplying

$f(x) = 1 + 2x + 0x^2 + 1x^3 + 0x^4 + 0x^5 + 1x^6 + 0x^7 \rightarrow$ as a list: 1, 2, 0, 1, 0, 0, 1, 0

$g(x) = 2 + 0x + 1x^2 + 0x^3 + 1x^4 + 0x^5 + 0x^6 + 1x^7 \rightarrow$ as a list: 2, 0, 1, 0, 1, 0, 0, 1

The goal: compute $f(x)$ times $g(x)$, reduced mod $(x^8 + 1)$ and mod 17 - and get the exact same answer whether we do it the slow schoolbook way or the fast NTT way.

Why we transform f and g at all

Multiplying two 8-coefficient polynomials the normal (schoolbook) way means checking every pair of terms against each other - a real amount of work. NTT rearranges each polynomial's numbers into a different list first; once both lists are rearranged this way, multiplying them together becomes simple - just matching pairs straight across.

Step 1: Transform f (NTT)

We take numbers two at a time and combine them with one multiply, then one add and one subtract - this is the 'butterfly' step. There are two rounds; round 2 splits into two smaller groups, each using a different magic number.

Round 1 - magic number 13

Pair (position 0, position 4): values 1, 0

$$13 \times 0 = 0$$

$$1 + 0 = 1 \rightarrow \text{position 0}$$

$$1 - 0 = 1 \rightarrow \text{position 4}$$

Pair (position 1, position 5): values 2, 0

$$13 \times 0 = 0$$

$$2 + 0 = 2 \rightarrow \text{position 1}$$

$$2 - 0 = 2 \rightarrow \text{position 5}$$

Pair (position 2, position 6): values 0, 1

$$13 \times 1 = 13$$

$$0 + 13 = 13 \rightarrow \text{position 2}$$

$$0 - 13 = -13 \rightarrow 4 \pmod{17} \rightarrow \text{position 6}$$

Pair (position 3, position 7): values 1, 0

$$13 \times 0 = 0$$

$$1 + 0 = 1 \rightarrow \text{position 3}$$

$$1 - 0 = 1 \rightarrow \text{position 7}$$

List after Round 1: 1, 2, 13, 1, 1, 2, 4, 1

Round 2, Group A - magic number 9 (positions 0-3)

Pair (position 0, position 2): values 1, 13

$$9 \times 13 = 117 \rightarrow 15 \pmod{17}$$

$$1 + 15 = 16 \rightarrow \text{position 0}$$

$$1 - 15 = -14 \rightarrow 3 \pmod{17} \rightarrow \text{position 2}$$

Pair (position 1, position 3): values 2, 1

$$9 \times 1 = 9$$

$$2 + 9 = 11 \rightarrow \text{position 1}$$

$$2 - 9 = -7 \rightarrow 10 \pmod{17} \rightarrow \text{position 3}$$

Round 2, Group B - magic number 15 (positions 4-7)

Pair (position 4, position 6): values 1, 4

$$15 \times 4 = 60 \rightarrow 9 \pmod{17}$$

$$1 + 9 = 10 \rightarrow \text{position 4}$$

$$1 - 9 = -8 \rightarrow 9 \pmod{17} \rightarrow \text{position 6}$$

Pair (position 5, position 7): values 2, 1

$$15 \times 1 = 15$$

$$2 + 15 = 17 \rightarrow 0 \pmod{17} \rightarrow \text{position 5}$$

$$2 - 15 = -13 \rightarrow 4 \pmod{17} \rightarrow \text{position 7}$$

Final transformed f (called $\hat{f}$): 16, 11, 3, 10, 10, 0, 9, 4

Step 2: Transform g (NTT)

Exact same process as Step 1, run on g instead of f.

Round 1, magic number 13

Pair (position 0, position 4): values 2, 1

$$13 \times 1 = 13$$

$$2 + 13 = 15 \rightarrow \text{position 0}$$

$$2 - 13 = -11 \rightarrow 6 \pmod{17} \rightarrow \text{position 4}$$

Pair (position 1, position 5): values 0, 0

$$13 \times 0 = 0$$

$$0 + 0 = 0 \rightarrow \text{position 1}$$

$$0 - 0 = 0 \rightarrow \text{position 5}$$

Pair (position 2, position 6): values 1, 0

$$13 \times 0 = 0$$

$$1 + 0 = 1 \rightarrow \text{position 2}$$

$$1 - 0 = 1 \rightarrow \text{position 6}$$

Pair (position 3, position 7): values 0, 1

$$13 \times 1 = 13$$

$$0 + 13 = 13 \rightarrow \text{position 3}$$

$$0 - 13 = -13 \rightarrow 4 \pmod{17} \rightarrow \text{position 7}$$

Round 2, Group A - magic number 9 (positions 0-3, using Round 1 results)

Pair (position 0, position 2): values 15, 1

$$9 \times 1 = 9$$

$$15 + 9 = 24 \rightarrow 7 \pmod{17} \rightarrow \text{position 0}$$

$$15 - 9 = 6 \rightarrow \text{position 2}$$

Pair (position 1, position 3): values 0, 13

$$9 \times 13 = 117 \rightarrow 15 \pmod{17}$$

$$0 + 15 = 15 \rightarrow \text{position 1}$$

$$0 - 15 = -15 \rightarrow 2 \pmod{17} \rightarrow \text{position 3}$$

Round 2, Group B - magic number 15 (positions 4-7)

Pair (position 4, position 6): values 6, 1

$$15 \times 1 = 15$$

$$6 + 15 = 21 \rightarrow 4 \pmod{17} \rightarrow \text{position 4}$$

$$6 - 15 = -9 \rightarrow 8 \pmod{17} \rightarrow \text{position 6}$$

Pair (position 5, position 7): values 0, 4

$$15 \times 4 = 60 \rightarrow 9 \pmod{17}$$

$$0 + 9 = 9 \rightarrow \text{position 5}$$

$$0 - 9 = -9 \rightarrow 8 \pmod{17} \rightarrow \text{position 7}$$

Final transformed g (called $\hat{g}$): 7, 15, 6, 2, 4, 9, 8, 8

Step 3: Multiply the transformed lists

f_hat and g_hat get multiplied 2 numbers at a time (4 pairs total, since we have 8 numbers). Each pair also needs its own magic number (called gamma here): 9, 8, 15, 2 for pairs 1-4. The formula for each pair (a_0, a_1) from f_hat and (b_0, b_1) from g_hat :

$$c_0 = (a_0 \times b_0) + (a_1 \times b_1 \times \text{gamma}) \quad c_1 = (a_0 \times b_1) + (a_1 \times b_0)$$

Pair 1: $f_hat=(16,11)$, $g_hat=(7,15)$, $\text{gamma}=9$

$$\begin{aligned} c_0 &= (16 \times 7) + (11 \times 15 \times 9) = 112 + 1485 \\ 112 \bmod 17 &= 10, \quad 1485 \bmod 17: 11 \times 15 = 165, 165 \bmod 17 = 12, 12 \times 9 = 108, 108 \bmod 17 = 6 \\ c_0 &= 10 + 6 = 16 \\ c_1 &= (16 \times 15) + (11 \times 7) = 240 + 77 \\ 240 \bmod 17 &= 2, \quad 77 \bmod 17 = 9 \\ c_1 &= 2 + 9 = 11 \\ \text{result: } &16, 11 \end{aligned}$$

Pair 2: $f_hat=(3,10)$, $g_hat=(6,2)$, $\text{gamma}=8$

$$\begin{aligned} c_0 &= (3 \times 6) + (10 \times 2 \times 8) = 18 + 160 \\ 18 \bmod 17 &= 1, \quad 160 \bmod 17 = 7 \\ c_0 &= 1 + 7 = 8 \\ c_1 &= (3 \times 2) + (10 \times 6) = 6 + 60 \\ 60 \bmod 17 &= 9 \\ c_1 &= 6 + 9 = 15 \\ \text{result: } &8, 15 \end{aligned}$$

Pair 3: $f_hat=(10,0)$, $g_hat=(4,9)$, $\text{gamma}=15$

$$\begin{aligned} c_0 &= (10 \times 4) + (0 \times 9 \times 15) = 40 + 0 = 40 \rightarrow 6 \pmod{17} \\ c_1 &= (10 \times 9) + (0 \times 4) = 90 + 0 = 90 \rightarrow 5 \pmod{17} \\ \text{result: } &6, 5 \end{aligned}$$

Pair 4: $f_hat=(9,4)$, $g_hat=(8,8)$, $\text{gamma}=2$

$$\begin{aligned} c_0 &= (9 \times 8) + (4 \times 8 \times 2) = 72 + 64 \\ 72 \bmod 17 &= 4, \quad 64 \bmod 17 = 13 \\ c_0 &= 4 + 13 = 17 \rightarrow 0 \pmod{17} \\ c_1 &= (9 \times 8) + (4 \times 8) = 72 + 32 \\ 72 \bmod 17 &= 4, \quad 32 \bmod 17 = 15 \\ c_1 &= 4 + 15 = 19 \rightarrow 2 \pmod{17} \\ \text{result: } &0, 2 \end{aligned}$$

Combined result (called h_hat): 16, 11, 8, 15, 6, 5, 0, 2

Step 4: Transform back (inverse NTT)

$\hat{h}$ is still in the 'transformed' shape - not real coefficients yet. This step undoes the transform from Steps 1/2, run backwards, using the same magic numbers in reverse order, then one final scaling step at the end.

Round 1 (reverse), magic number 15 (positions 0-3)

Pair (position 0, position 2): values 16, 8

$$16 + 8 = 24 \rightarrow 7 \pmod{17} \rightarrow \text{position 0}$$

$$15 \times (8-16) = 15 \times -8 = -120 \rightarrow 16 \pmod{17} \rightarrow \text{position 2}$$

Pair (position 1, position 3): values 11, 15

$$11 + 15 = 26 \rightarrow 9 \pmod{17} \rightarrow \text{position 1}$$

$$15 \times (15-11) = 15 \times 4 = 60 \rightarrow 9 \pmod{17} \rightarrow \text{position 3}$$

Round 1 (reverse), magic number 9 (positions 4-7)

Pair (position 4, position 6): values 6, 0

$$6 + 0 = 6 \rightarrow \text{position 4}$$

$$9 \times (0-6) = 9 \times -6 = -54 \rightarrow 14 \pmod{17} \rightarrow \text{position 6}$$

Pair (position 5, position 7): values 5, 2

$$5 + 2 = 7 \rightarrow \text{position 5}$$

$$9 \times (2-5) = 9 \times -3 = -27 \rightarrow 7 \pmod{17} \rightarrow \text{position 7}$$

Round 2 (reverse), magic number 13 - all 4 pairs, spanning the whole list

Pair (position 0, position 4): values 7, 6

$$7 + 6 = 13 \rightarrow \text{position 0}$$

$$13 \times (6-7) = 13 \times -1 = -13 \rightarrow 4 \pmod{17} \rightarrow \text{position 4}$$

Pair (position 1, position 5): values 9, 7

$$9 + 7 = 16 \rightarrow \text{position 1}$$

$$13 \times (7-9) = 13 \times -2 = -26 \rightarrow 8 \pmod{17} \rightarrow \text{position 5}$$

Pair (position 2, position 6): values 16, 14

$$16 + 14 = 30 \rightarrow 13 \pmod{17} \rightarrow \text{position 2}$$

$$13 \times (14-16) = 13 \times -2 = -26 \rightarrow 8 \pmod{17} \rightarrow \text{position 6}$$

Pair (position 3, position 7): values 9, 7

$$9 + 7 = 16 \rightarrow \text{position 3}$$

$$13 \times (7-9) = 13 \times -2 = -26 \rightarrow 8 \pmod{17} \rightarrow \text{position 7}$$

List before the final step: 13, 16, 13, 16, 4, 8, 8, 8

Final scaling step

Multiply every single number in the list by 13 (this undoes a leftover factor of 4 from the transform - $4 \times 13 = 52$, and $52 \pmod{17} = 1$, so 13 is chosen specifically to cancel it out).

$$13 \times 13 = 169 \rightarrow 16 \quad 16 \times 13 = 208 \rightarrow 4 \quad 13 \times 13 = 169 \rightarrow 16 \quad 16 \times 13 = 208 \rightarrow 4$$

$$4 \times 13 = 52 \rightarrow 1 \quad 8 \times 13 = 104 \rightarrow 2 \quad 8 \times 13 = 104 \rightarrow 2 \quad 8 \times 13 = 104 \rightarrow 2$$

Final result: 16, 4, 16, 4, 1, 2, 2, 2

Step 5: Check against the slow schoolbook method

Multiplying f and g the normal (schoolbook) way, reduced mod (x^8+1) and mod 17, gives the same list:

Schoolbook result: 16, 4, 16, 4, 1, 2, 2, 2

NTT result: 16, 4, 16, 4, 1, 2, 2, 2

MATCH - both methods give the exact same answer.

How this maps onto real FIPS 203 / Kyber

Everything above follows the exact same structure as FIPS 203's real algorithms:

Step 1/2 = Algorithm 9 (NTT). Step 3 = Algorithms 11/12 (MultiplyNTTs, BaseCaseMultiply).

Step 4 = Algorithm 10 (NTT-1). The only difference at real Kyber scale ($n=256$, $q=3329$, $\text{zeta}=17$) is more numbers and more rounds (8 rounds instead of 2) - the actual arithmetic being done at each step is identical to what's shown here.